IaaS vs PaaS vs SaaS

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1. Conceptual Layering (Who Manages What)

Layer	On-Prem	IaaS	PaaS	SaaS
Application	You	You	You	Provider
Data	You	You	You	Provider
Runtime	You	You	Provider	Provider
Middleware	You	You	Provider	Provider
OS	You	You	Provider	Provider
Virtualization	You	Provider	Provider	Provider
Servers	You	Provider	Provider	Provider
Storage	You	Provider	Provider	Provider
Networking	You	Provider	Provider	Provider

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2. IaaS (Infrastructure as a Service)

Definition

Provides **raw computing infrastructure** (VMs, storage, networks).
You get maximum flexibility but also maximum responsibility.

Architecture Flow

User → Cloud Provider → Virtual Machine → OS → App

Key Features

- Virtual Machines (VMs)

- Custom OS installation
- Full control over environment
- Scalable infrastructure

Examples

- Microsoft Azure Virtual Machines
- Amazon Web Services EC2
- Google Cloud Compute Engine

Use Cases

- Lift-and-shift migration
- Custom software environments
- Big data clusters (manual Spark setup)

Advantages

- Maximum control
- Flexible configurations
- Supports legacy apps

Limitations

- Requires system admin knowledge
- Manual scaling (unless configured)
- Maintenance overhead

Example Scenario

You launch a VM and install:

- Linux
- Java

- Spark
  → You manage everything

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3. PaaS (Platform as a Service)

Definition

Provides a **ready-to-use platform** to build, run, and deploy applications.

Architecture Flow

User → Platform → Runtime → Managed Infrastructure

Key Features

- Managed runtime (Java, Python, Node)
- Auto scaling
- Built-in deployment tools
- CI/CD integration

Examples

- Microsoft Azure App Service
- Google Cloud App Engine
- Salesforce Platform

Use Cases

- Web apps
- APIs
- Backend services
- Data pipelines (Databricks, Synapse)

Advantages

- Faster development
- No OS management
- Built-in scalability

Limitations

- Less control
- Vendor lock-in risk
- Limited customization

Example Scenario

You upload code:

```
def app():
  return "Hello"
```

→ Platform handles:

- Deployment
- Scaling
- Runtime

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4. SaaS (Software as a Service)

Definition

Ready-to-use **software accessed via browser or API**.

Architecture Flow

User → Internet → Software Application

Key Features

- No installation
- Subscription-based
- Accessible anywhere

Examples

- Gmail
- Microsoft 365
- Google Docs

Use Cases

- Email
- CRM (Salesforce)
- Collaboration tools

Advantages

- Zero maintenance
- Instant access
- Highly scalable

Limitations

- Minimal control
- Data dependency on provider
- Customization limits

Example Scenario

You open Gmail → no setup, just use.

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5. Deep Comparison

Feature	IaaS	PaaS	SaaS
Control	High	Medium	Low
Responsibility	High	Medium	Low
Setup Effort	High	Medium	None
Scaling	Manual/Config	Automatic	Automatic
Flexibility	High	Medium	Low
Cost	Pay infra	Pay platform	Subscription
Best For	Infra control	Development	End users

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6. Key Insight

- **IaaS → You manage most**
- **PaaS → You focus on code**
- **SaaS → You just use the product**